

Note: Please reframe the answers as per your understanding

METEORLOGY

Meteorology is the study of the Earth's atmosphere and the physical processes that occur within it.

Q: Why meteorology is important?

A: Meteorology is vital for pilots as it helps them **anticipate and navigate through changing weather** conditions, ensuring flight safety.

Pilots rely on meteorological forecasts to plan routes, avoid hazardous weather, and make informed decisions during flight. Understanding meteorology enables pilots to interpret weather-related data and optimize fuel efficiency and flight performance.

Overall, meteorology plays a crucial role in ensuring the safety and efficiency of flight operations.

About atmosphere

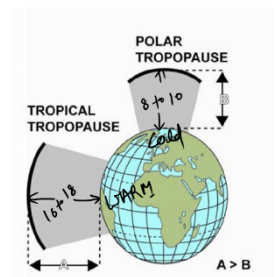
**Q: What are the standard conditions for temperature, pressure, and density at Mean Sea Level (MSL) in the International Standard Atmosphere (ISA)?

A: The standard conditions at Mean Sea Level (MSL) in the International Standard Atmosphere (ISA) are:

- Temperature: $+15^{\circ}\text{C}$
- Pressure: 1013.25 hPa or millibars (or 29.92 inches of mercury)
- Density: 1.225 kilograms per cubic meter or 1225 grams / cubic metre,
- A lapse rate of $0.65^{\circ}\text{C}/100\text{ m}$ ($1.98^{\circ}\text{C}/1000\text{ ft}$) up to 11 km (36 090 ft),
- A constant temperature of -56.5°C from 11 km to 20 km (65 617 ft),
- $g=9.81\text{ m/sec}^2$

*Q: At what elevation does the tropopause typically occur in the Earth's atmosphere?

A: The tropopause typically occurs at an altitude of around 11 km above the Earth's surface. But it varies from equator to pole, it higher over equator (16-18 km) but lower over poles(8-10km).



*Q: What is the altitude range of the stratosphere within the Earth's atmosphere?

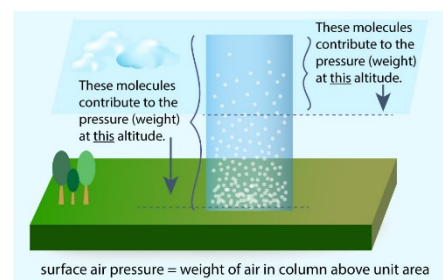
A: The stratosphere extends from approximately 11 to 50 kilometres above the Earth's surface. Lower part of the stratosphere (11 to 20 kilometres) is tropopause where temperature is constant after 20 km temperature starts increasing.

Atmospheric Layers Relevant to Aviation:

1. Troposphere: Temp decreases - Extends from the Earth's surface up to about 36,000 feet (11 km). Weather phenomena and most commercial airline operations occur in this layer.
2. Stratosphere: Temp increases due to presence of Ozone —11 to 50 km. Commercial supersonic flights and some weather balloons operate here.
3. Mesosphere: Temp decreases— 50 km to 85 km. Meteors burn up in this layer.
4. Thermosphere: Temp increases due to presence of ionised particles -- 80 to 700 km.
5. Exosphere: Temp decreases - above 700 km - Satellites orbit in this layer.

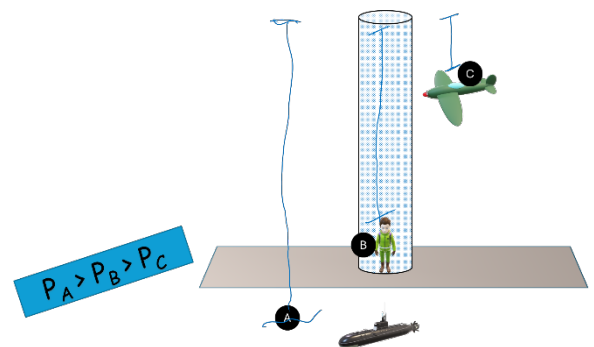
*Q: Define atmospheric pressure?

A: Atmospheric pressure is defined as the force exerted per unit area by the weight of the air above a given point in the Earth's atmosphere.



*Q: What causes a decrease in pressure with increasing altitude in the Earth's atmosphere?

A: Pressure decreases with increasing altitude due to the decreasing weight of the air column above. As altitude increases, there is less air above to exert less pressure downward, resulting in lower atmospheric pressure.



**Q. Why do aircraft use cabin pressurization?

Commercial aircraft fly best at high altitudes – that is a fact. This enables them to enhance fuel consumption efficiency and avoid potential severe weather and turbulence factors. However, for humans the situation is precisely the opposite. The higher we go, the less oxygen there is available to breathe.

This happens because air density decreases with altitude. Thus, air molecules spread out more, decreasing their density and – with that – there is less oxygen available for each breath of air. All this makes it increasingly harder to breathe for us.

At 18,000 feet, the amount of oxygen halves compared to what we normally have at sea level.

In fact, going much higher than 10,000 feet without the help of modern technology can cause altitude sickness, also known as hypoxia. Hypoxia can lead to dizziness, headache, difficulty thinking, unconsciousness and eventually death.

Thankfully, modern jet airliners are engineering miracles. Apart from getting us across the world safely in a matter of hours, they also act as a flying pressure chamber, controlling the air entering and exiting the pressurized cabin - air management systems – maintaining the necessary pressure so that passenger and crew can breathe comfortably.

Q. How high we can fly without a pressurised cabin?

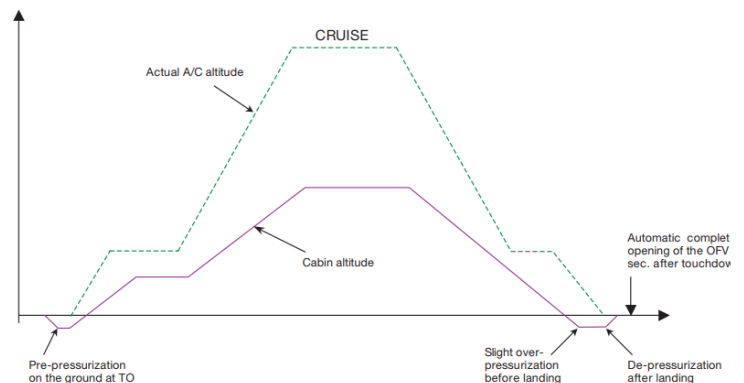
Most modern aircraft employ a cabin pressure control mechanism to maintain a safe and comfortable breathing environment for passengers & crew throughout the flight. **The general rule is that planes should have cabin pressurization when they go above 10,000.**

*Q. What is the cabin pressure on a plane?

For most of airliner Cabin altitude is approximately 8,000 feet (760 hPa).

Actually, the cabin pressure varies as per the altitude as shown in diagram.

The cabin pressurization system of the Airbus A320 is designed to maintain a maximum cabin altitude of 8,000ft (approximately 11.03psi) when the aircraft is cruising at high altitude by controlling the opening and closing of an outflow valve. Further, it is designed to limit the differential pressure between the cabin pressure and the ambient pressure to a maximum of 8.06psi.



*Q. What happens if you lose cabin pressure?

In the unlikely event of a loss of cabin pressure, oxygen masks become available for everyone onboard to help them breathe normally until the aircraft reaches a lower altitude and cabin pressure is restored.

Q: Hypoxia

(1) Hypoxia

The symptoms of lack of sufficient oxygen to human body cells or tissues are called “hypoxia”, and there are following relations:

<u>Altitude (ft)</u>	<u>Stage</u>	<u>Symptoms</u>
0–10,000	Indifferent	None, but visual sensitivity reduced at night.
10,000–15,000	Compensatory	Major symptoms may not appear due to compensatory mechanisms.
15,000–20,000	Disturbance	Hazards of visual disturbance and intellectual impairment, etc.
above 20,000	Critical	Danger of rapid loss of consciousness with little or no warning and loss of life.

Q: What factors contribute to the temperature differences between the poles and the equator?

A: The temperature differences between the poles and the equator are primarily due to variations in solar radiation received at different latitudes. The equator receives more heating, leading to warmer temperatures, while the poles receive less direct heat resulting in colder temperatures. (differential Heating by Sun)

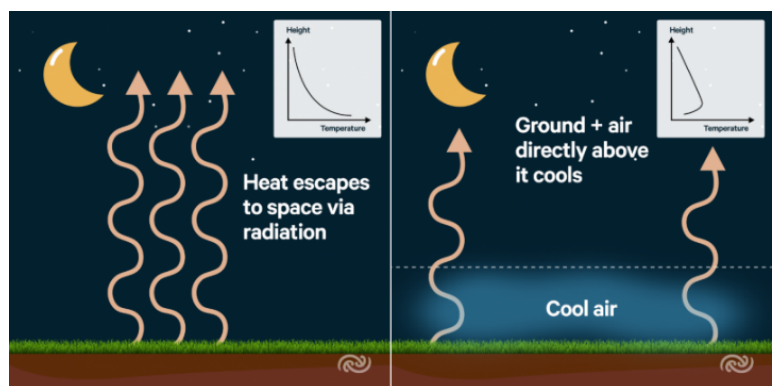


Fog

**Q: How Radiation fog forms?

A: Radiation fog forms during night and can be seen in early morning. During the day, the sun heats the Earth's surface, and after sun sets, this heat escapes into the atmosphere (i.e., nocturnal cooling).

On clear nights with little wind (5-7 kts), the ground loses heat rapidly, causing the air near the ground to cool.



As the air cools, its moisture condenses into tiny water droplets, forming fog. This fog covers the ground and reduces visibility, especially in valleys and low-lying areas.

Ideal Conditions for radiation fog formation: Clear skies (no cloud), calm winds, and high relative humidity are ideal conditions for radiation fog.

Aviation hazard:

- Radiation fog significantly reduces visibility, affecting take-offs, landings, and navigation. Pilots need to rely more on instruments for navigation and landing in foggy conditions.
- Airports may implement low-visibility procedures, affecting traffic flow and requiring specific pilot qualifications.
- Safety Precautions: Pilots must be trained to interpret weather forecasts and reports, understand the local climatology of their flight routes, and be prepared to alter plans as necessary.

Understanding the formation of radiation fog and its implications in climatology is essential for pilots to ensure safe and efficient flight operations, especially in terms of navigation and decision-making in response to weather conditions.

Q: How fog forms and what are the types of fog?

Fog is a surface-based cloud (restricting visibility) composed of either water droplets or ice crystals. Fog may form by cooling the air to its dew point or by adding moisture to the air near the ground. A small temperature/dew point spread is essential to the formation of fog. An abundance of condensation nuclei from combustion products makes fog prevalent in industrial areas.

Fog is classified by the way it is formed:

Radiation fog (ground fog) is formed when terrestrial radiation cools the ground, which in turn cools the air in contact with it. When the air is cooled to its dew point (or within a few degrees), fog will form. This fog will form most readily in warm, moist air over low, flatland areas on clear, calm (no wind) nights.

Advection fog (sea fog) is formed when warm, moist air moves (wind is required) over colder ground or water (e.g., an air mass moving inland from the coast in winter).

Upslope/orography fog is formed when moist, stable air is cooled to its dew point as it moves (wind is required) upsloping terrain. Cooling will be at the dry adiabatic lapse rate of approximately 3°C per 1,000 feet.

Precipitation (rain or drizzle)-induced fog is most commonly associated with frontal activity and is formed by relatively warm drizzle or rain falling through cooler air. Evaporation from the precipitation saturates the cool air and fog forms. This fog is especially critical because it occurs in the proximity of precipitation and other possible hazards such as icing, turbulence, and thunderstorms.

Steaming fog forms in the winter when cold, dry air passes from land areas over comparatively warm ocean waters in specially in higher latitude & cold region.

Cloud

*Q: How cloud forms?

A: Cloud formation begins when warm air rises and cools as it ascends through the atmosphere. As the air cools, it reaches its dew point—the temperature at which it becomes saturated with water vapor. At this point, the water vapor condenses onto tiny particles in the air, such as dust or aerosols, forming tiny droplets. These droplets cluster together to form visible clouds.

The process of cloud formation is crucial for pilots, as clouds can indicate weather conditions and affect flight visibility and turbulence. Understanding cloud formation helps pilots anticipate and navigate through different atmospheric conditions for safe flying.

**Q: Types of cloud?

Stability determines which of two types of clouds will be formed: cumuliform or stratiform.

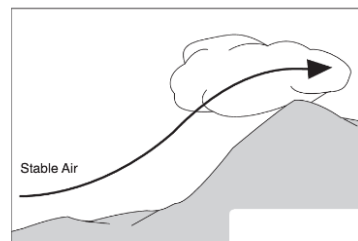
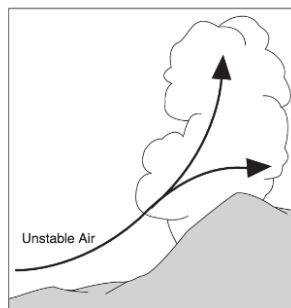
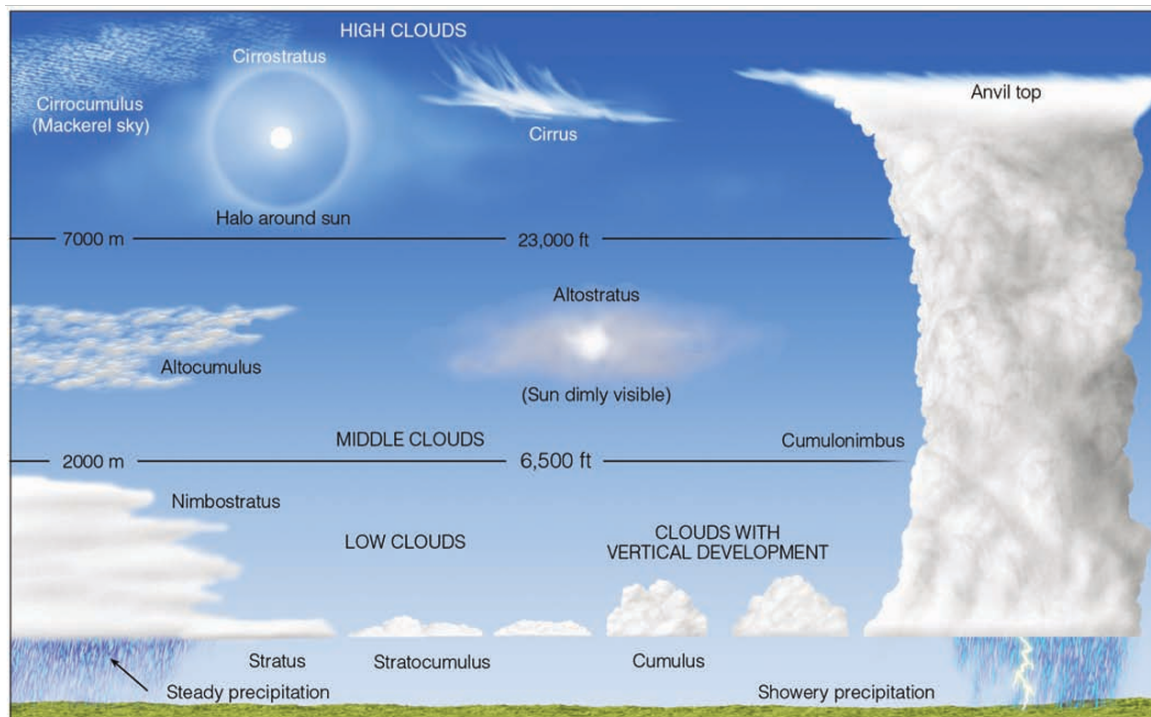


Figure 6-9. Stratiform clouds



Airframe Icing

The two primary types of icing are airframe icing and engine icing.

Airframe icing occurs on an aircraft, whenever supercooled water droplets (0 to -40 C) makes contact with any part of the aircraft that at temperature below freezing temp (Zero).

***Q: What are the type of airframe icing and how it affects the aircraft?**

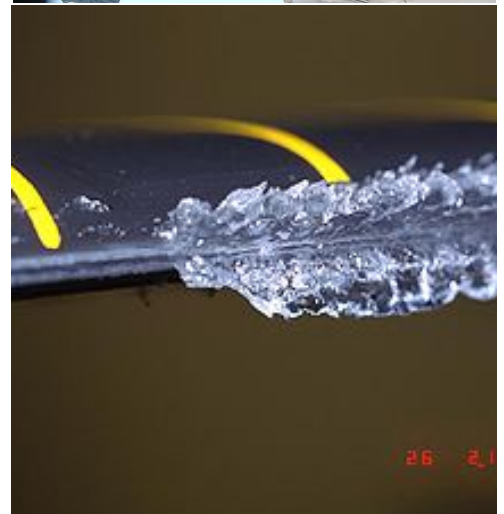
Rime Ice:

- **Temperature Range:** Typically forms at temperatures between -15 to -40°C.
- **Formation:** Occurs due to the rapid freezing of small water droplets.
- **Location:** Commonly found on the leading surfaces or inlet of the engine.
- **Characteristics:** Rime ice is granular, resembling rough sandpaper. It's brittle and generally easier to remove than clear ice.
- **Appearance:** Has a milky white appearance due to trapped air.



Glaze Ice (Clear Ice):

- **Temperature Range:** Forms at temperatures ranging from 0 to -20°C.
- **Formation:** Results from the slow freezing of large water droplets.
- **Characteristics:** Clear ice is hard, heavy, and tenacious, making its removal by de-icing equipment particularly difficult.
- **Appearance:** Transparent and smooth, which makes it difficult to detect and a hazard in aviation.



Both types of ice pose significant risks to aircraft, but they differ mainly in their formation process, appearance, and the impact they have on the surfaces they form on.

Hoar frost

A deposit of white ice crystals resembling ground frost, occurring in clear air when the airframe temperature is below 0°C and the ambient air reaches saturation, causing water vapor to sublimate directly into ice without becoming liquid.

It can form on the ground, often overnight, and must be cleared prior to take-off to avoid increased take-off run due to skin friction, obscured windscreens, and radio interference from ice on aerials.

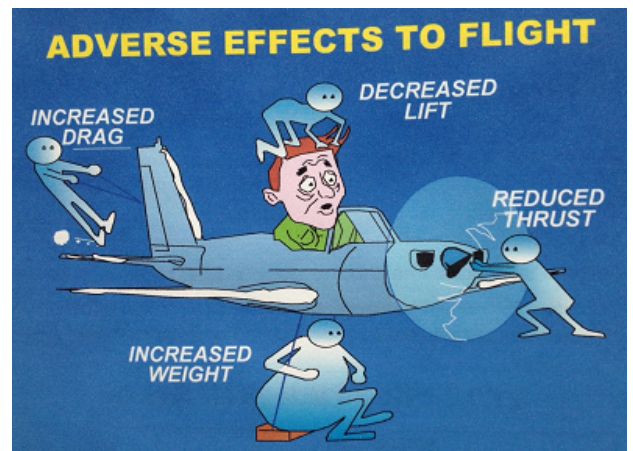
In flight, hoar frost can form during rapid descents from cold to warm moist air or when climbing through an inversion from below freezing temperatures.



The Hoarfrost is usually **not severe** and can be mitigated by flying into warmer air or increasing speed for kinetic heating, and it dissipates as the aircraft skins warms up due to frictional heating.

****Effect of Airframe Icing on Aircraft**

- **Increases weight.**
- Increases stalling speed.
- Increases fuel consumption.
- Ice formation disrupts aerodynamic shape, particularly at leading edges, Reduces lift. Increases drag.
- **Can obstruct pressure heads, leading to erroneous Pressure instrument readings.**
- **Uneven ice on propeller blades can induce severe engine vibration. Ice shedding from propellers can damage aircraft skin.**



Thunderstorm

Condition for formation of thunderstorm.

Thunderstorms occur in well-developed Cumulonimbus (Cb), though not all Cb's produce thunderstorms. They are most likely to occur when there is:

1. **A lapse rate greater than the SALR (unstable conditions)** through a layer at least 10,000 ft thick and extending above the freezing level.
2. **Sufficient water vapour** to form and maintain the cloud.
3. **Trigger action** to produce early saturation, thus enhancing instability.



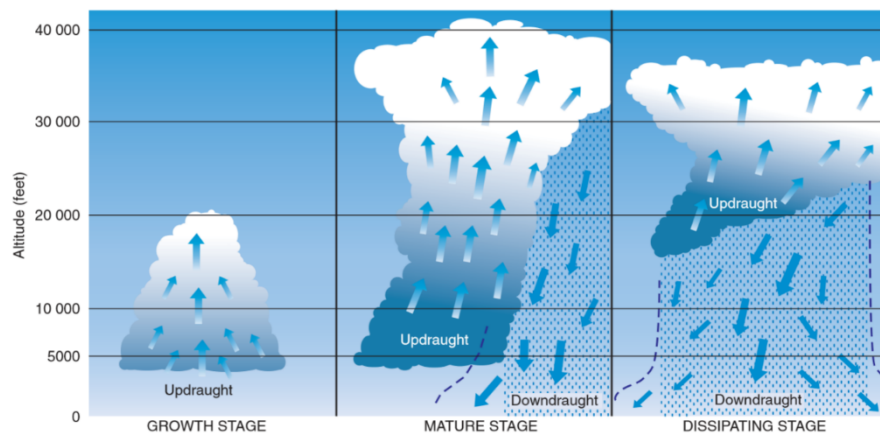
Which are

Convection	Upward movement of air that is warmer than its surroundings.
Orographic Uplift	Air is lifted as it moves over rising terrain, such as hills or mountains.
Convergence	Air flows coming from different directions meet, forcing the air upward.
Frontal Uplift	Air is forced upward at the boundary between two different air masses.

Thunderstorms are classified as:

Heat, or airmass type	Frontal type
more common in summer afternoon over land Formed by day, clear by night.	Most frequent in winters Forms over land or sea, day or night Usually formed in a line at a cold front or Cold occlusion

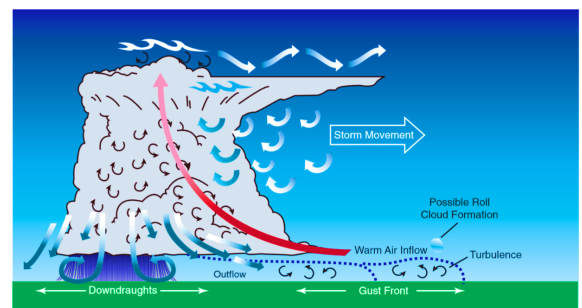
Life cycle of thunderstorm



Aviation Hazards:

I. Gust Fronts:

- Cold air mass from thunderstorm downdraft.
- Can extend 24-32 km and reach 6,000 ft high.
- Causes windshear, often invisible on radar.



II. Turbulence & Wind Shear:

- Moderate to severe turbulence near clouds.
- Hazardous during take-off and landing.
- Can cause airframe stress and unreliable instrument readings.

III. Microburst:

- Strong downdrafts from CB clouds.
- Speeds up to 60 kt, <5 km across, last 1-5 minutes.



IV. Hail:

— Damages aircraft structure.

- Likely in or near cloud anvils.
- Hail size increases with stronger updrafts.

V. Icing:

- Airframe icing from 0°C to -40°C.
- Severe icing near cloud bases (-7°C).
- Carburetor icing from -10°C to +30°C.

VI. Static:

- Interferes with radio equipment.
- St. Elmo's Fire indicates charged atmosphere.



VII. Lightning:

- Occurs $\pm 10^\circ\text{C}$ of freezing level.
- Causes temporary blindness, instrument interference, and airframe damage.



VIII. Water Ingestion:

- High water concentration can damage engines.
- Affects pitot heads.

IX. Tornadoes:

- Arise from intense convergence.
- Characterized by rotating air column.
- Avoidance criteria include maintaining vertical separation and distance from storm clouds.



General Guidelines:

- Maintain 5,000 ft vertical separation from storm clouds.
- Keep 10nm distance from tall, rapidly growing clouds.
- Avoid flying beneath cumulonimbus cloud overhangs.

****Q: Explain the phenomenon of wind shear and its impact on flight operations.**

Wind shear refers to a sudden change in wind speed and/or direction over a short distance, either horizontally or vertically.

1. Types of Wind Shear:

- Vertical Wind Shear: Changes in wind speed or direction with altitude.
- Horizontal Wind Shear: Changes in wind speed or direction over a horizontal distance.

2. Causes:

- Frontal Systems: Wind direction and speed can change dramatically around weather fronts.
- Thunderstorms: Downbursts and microbursts from thunderstorms can create strong wind shears.
- Temperature Inversions: Can cause changes in wind speed and direction near the ground.
- Jet Streams: High-altitude, fast-flowing air currents can cause wind shear.

Impact on Flight Operations:

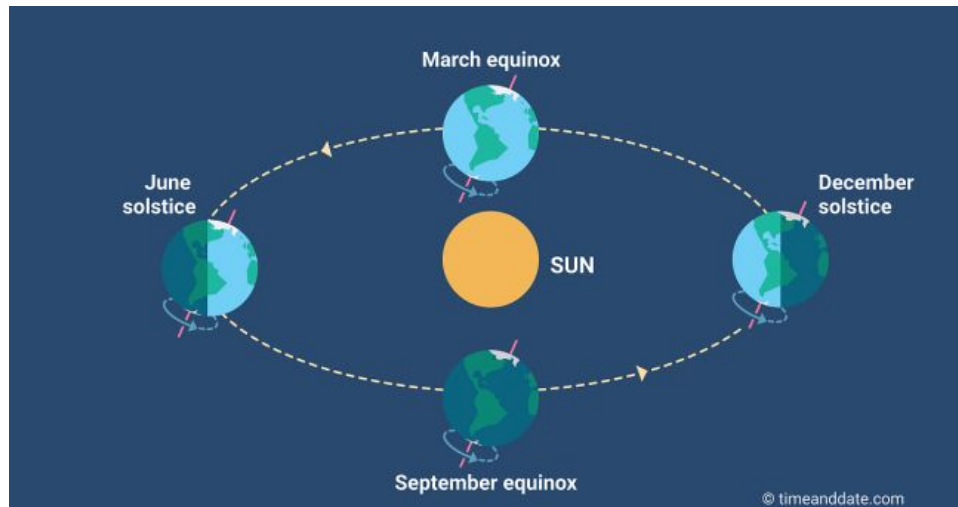
- Turbulence: Wind shear can create turbulence, affecting passenger comfort and aircraft control.
- Take-off and Landing: A sudden windshear (may be due to microburst, Subsidence) when aircraft is closer to the ground can be hazardous. just after the take-off climb may be suppressed and aircraft may start sinking altitude or approach can be hazardous.
- Altitude Changes: Rapid changes in wind speed/direction can cause airspeed fluctuations, potentially leading to loss of control.
- Flight Path Deviations: Pilots may need to adjust their flight paths to avoid areas of known or suspected wind shear.

Mitigation and Safety Measures:

- Training: Pilots receive training on recognizing and reacting to wind shear conditions.
- Technology: Aircraft are equipped with wind shear detection systems.
- Communication: Air traffic control and pilots share information about wind shear reports.

Understanding wind shear is crucial for pilots as it directly affects aircraft performance, safety, and decision-making during critical phases of flight. Proper training, technology, and communication are essential to mitigate the risks associated with wind shear.

Seasons in India



****Q: which all seasons we experience in India**

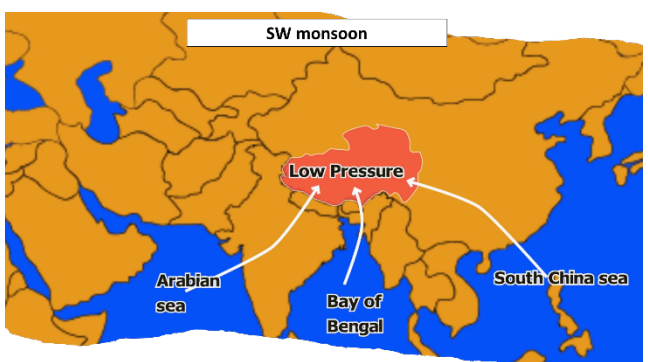
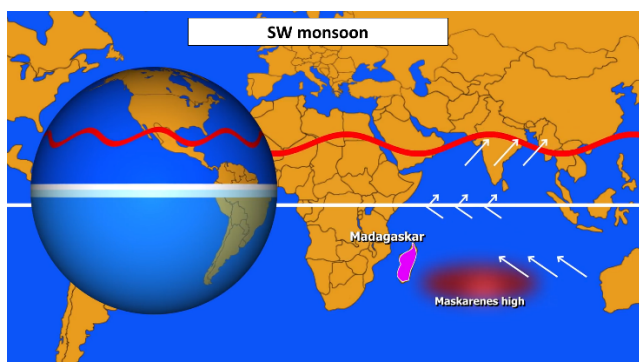
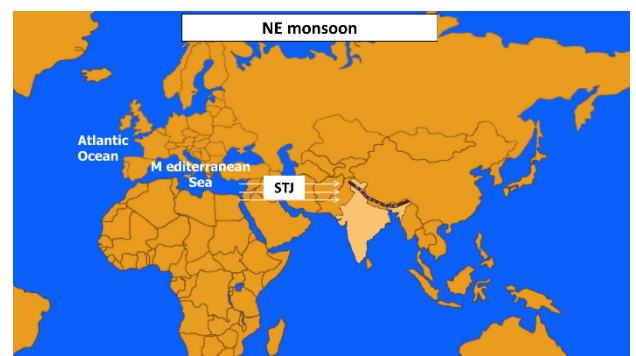
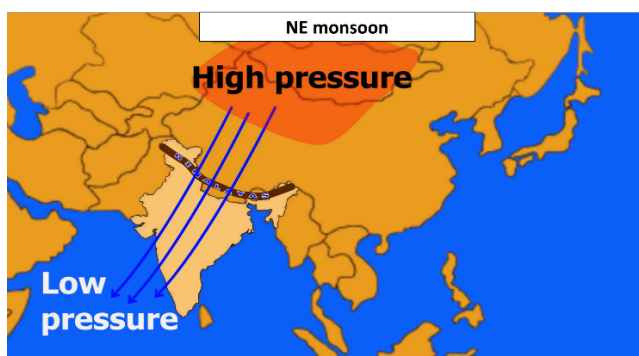
On the basis of climate, the period of year has been divided into four seasons in India.

N.E MONSOON (Winter) – December to February

PRE-MONSOON (Summer) – March to May

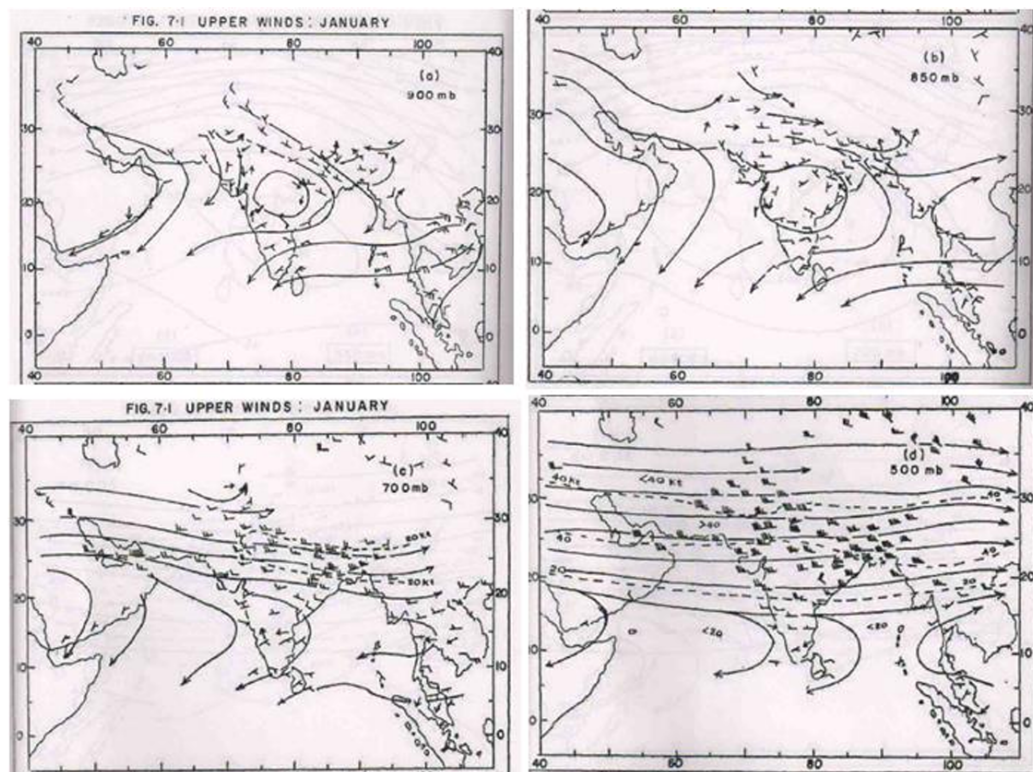
S.W MONSOON (Rainy season) – June to September

POST MONSOON (Transition) – October to November



N.E MONSOON (Winter) – December to February

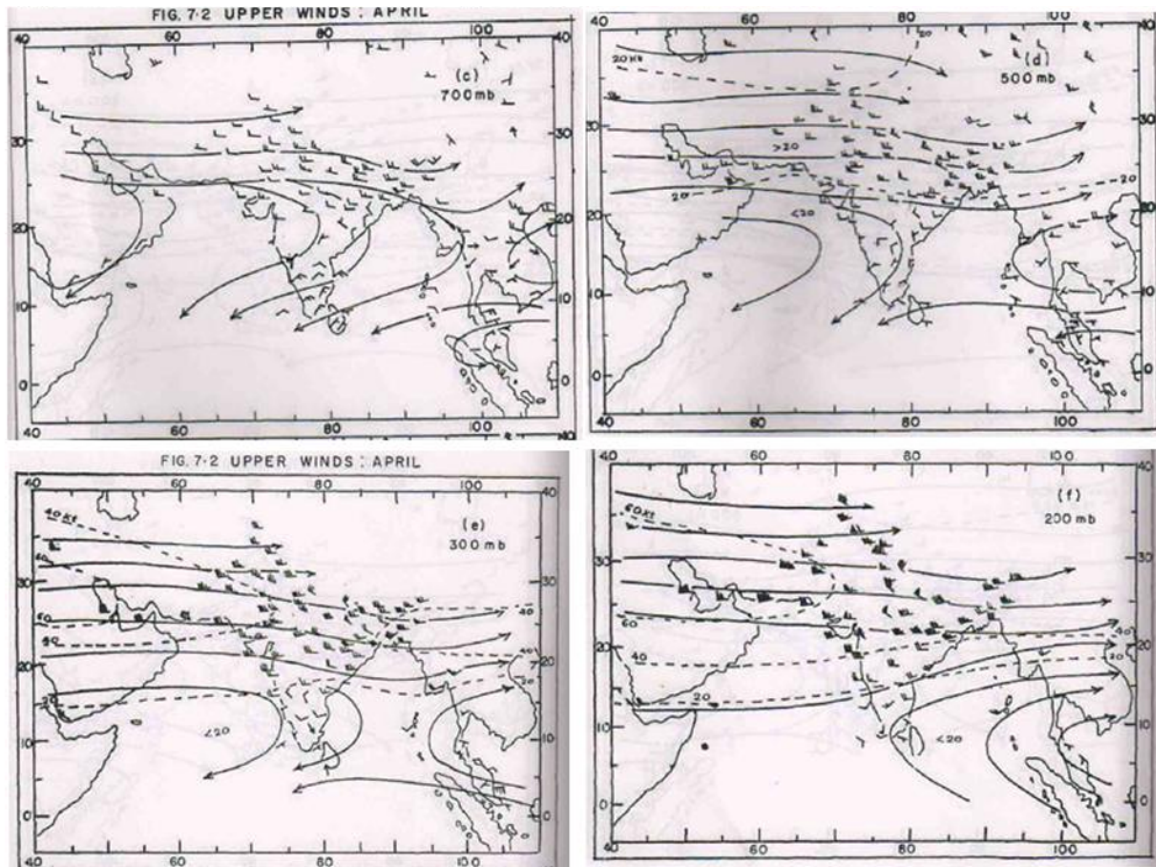
- High pressure (1012-1019 hPa) over North India (Rajasthan, Delhi, Punjab, Pakistan) and Low pressure over the Indian ocean.
- Low-temperature Low humidity (cold air is less humid)
- Cold waves are common in Jan -Feb in J&K, Rajasthan, Gujrat, and Punjab.
- Fog, mist and haze are common all over the country.
- Poor visibility is main aviation hazard, due Widespread Radiation fog and cold waves mainly in N India.
- Advection fog occurs over the coastal area.
- CAT in the vicinity of Subtropical Jet stream and Mountain waves over higher terrain areas.
- The freezing level gets lower in winter so Icing can be experienced at a low level.
- Rainfall is very less all over India, except in J&K.
- Rainfall over northern part of India (Kashmir, Punjab, Haryana, Himachal Pradesh, Rajasthan, West Uttar Pradesh) occurs in association with the passage of western disturbances and induced systems, sometimes extending in Northeastern states till Assam-Arunachal Pradesh.
- Rainfall in the south peninsula due to the movement of easterly waves.
- The STWJ is generally observed over North India from October to May and is strongest during the winter.



PRE-MONSOON (Summer) – March to May

This is the transition phase when the winter pattern starts to change in the monsoon pattern.

- The temperature starts rising, pressure falling. Weak pressure gradient over the country (1006 and 1008 hPa)
- Weak low over Pakistan Rajasthan and exceeding trough line over East UP-Bihar.
- Shallow high pressure in Arabian Sea and Bay of Bengal.
- Heat wave & ANDHI (Blinding DUST Storm) in NW part of India over plain (i.e., Punjab, Haryana, Rajasthan) and adjoining West Uttar Pradesh.
- Thunderstorms over NE India (i.e., Assam and adjacent states, West Bengal and Sikkim, Odisha and Bihar).
- Thunderstorms over NE India are known as 'Norwesters' (moving from NW direction) or 'Kal-baishakhi' (Highly destructive local storms in the month of Baishakh).
- Tropical Revolving Storm forms in the Arabian Sea and Bay of Bengal
- Bay of Bengal is more prone to depressions and storms than the Arabian Sea in the pre-monsoon season. Peak storm activity normally occurs during the month of May. Also, during this month most of the storms attain severe intensity.
- The major rainfall areas during the season are Assam and adjacent states, Kashmir (20-40 cm), Himachal Pradesh and south Kerala (10-20 cm). In the northeastern states the rainfall amount is higher (100 cm or more) in the southern parts of Meghalaya.



S.W MONSOON (Rainy season) – June to September

July may be taken as the representative month of the monsoon season.

SW monsoon or summer Monsoon can be considered as sea breeze but larger scale.

- An intense seasonal low-pressure (994hPa) is over Pakistan and adjoining Rajasthan. An anticyclone appears at 500 hPa and aloft over the Tibetan Plateau.
- High pressure will be near Madagascar and Australia. (Mascarenes High)
- Pressure Gradient is maximum during this season & the country receives 80% of annual rainfall during this season.
- SW monsoon is the principal rainy season in India except in Jammu and Kashmir and Tamil Nadu (TN gets rainfall in Oct and Nov).

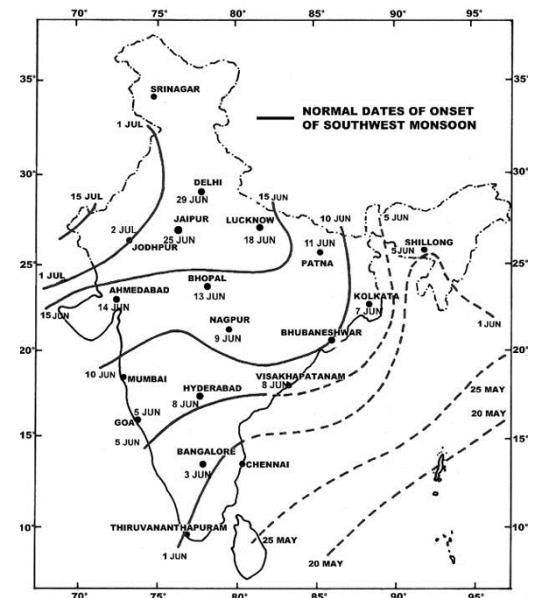
monsoon advances over the country in two branches.

The Arabian Sea branch which normally sets in monsoon rainfall over Kerala on June 1 advances rapidly northward along the west coast and reaches Gujarat by June 15. This branch brings monsoon rainfall over the entire Peninsula and central India.

Simultaneously the Bay branch for monsoon advances northward and enters Bangladesh, Assam and adjacent states by June 5. The topography and orientation of the hills of Assam and Meghalaya obstruct the current in the north and east resulting the deflection of the current towards west under the influence of the Himalayas.

The bay branch thus enters over the Gangetic plains of north India as a Southeasterly/Easterly current. Both the branches meet over north India around longitude 80° E.

The withdrawal of monsoon commences from western parts of west Rajasthan by Sept 1. It withdraws from NW India by Sept 15 and the withdrawal is complete from the country outside Tamil Nadu and Kerala by Oct 15 when NE monsoon normally sets in over Tamil Nadu; Southern part of India and Kerala experience rainfall during post monsoon season.



- Precipitation during the monsoon could be in the form of thundershowers, showers, or rain. Orography around India influence monsoon circulation and thus the spatial distribution of rainfall in the season.

Hills and mountains get more rainfall than the neighbouring plains. Mawsynram (1401 m) in Meghalaya experience the season's highest rainfall followed by Cherrapunji (835 cm), also in Meghalaya. over western ghats and adjoining west coast between 13° N and 20° N where it is 250 cm or more. Coastal Karnataka rainfall during the season exceeds 300 cm. Around 250 cm rainfall occurs in the southern parts of Meghalaya and in Mizoram. 100 – 150 cm over NE India, Madhya Pradesh, Vidarbha, Hills of West U.P. HP and Telangana.

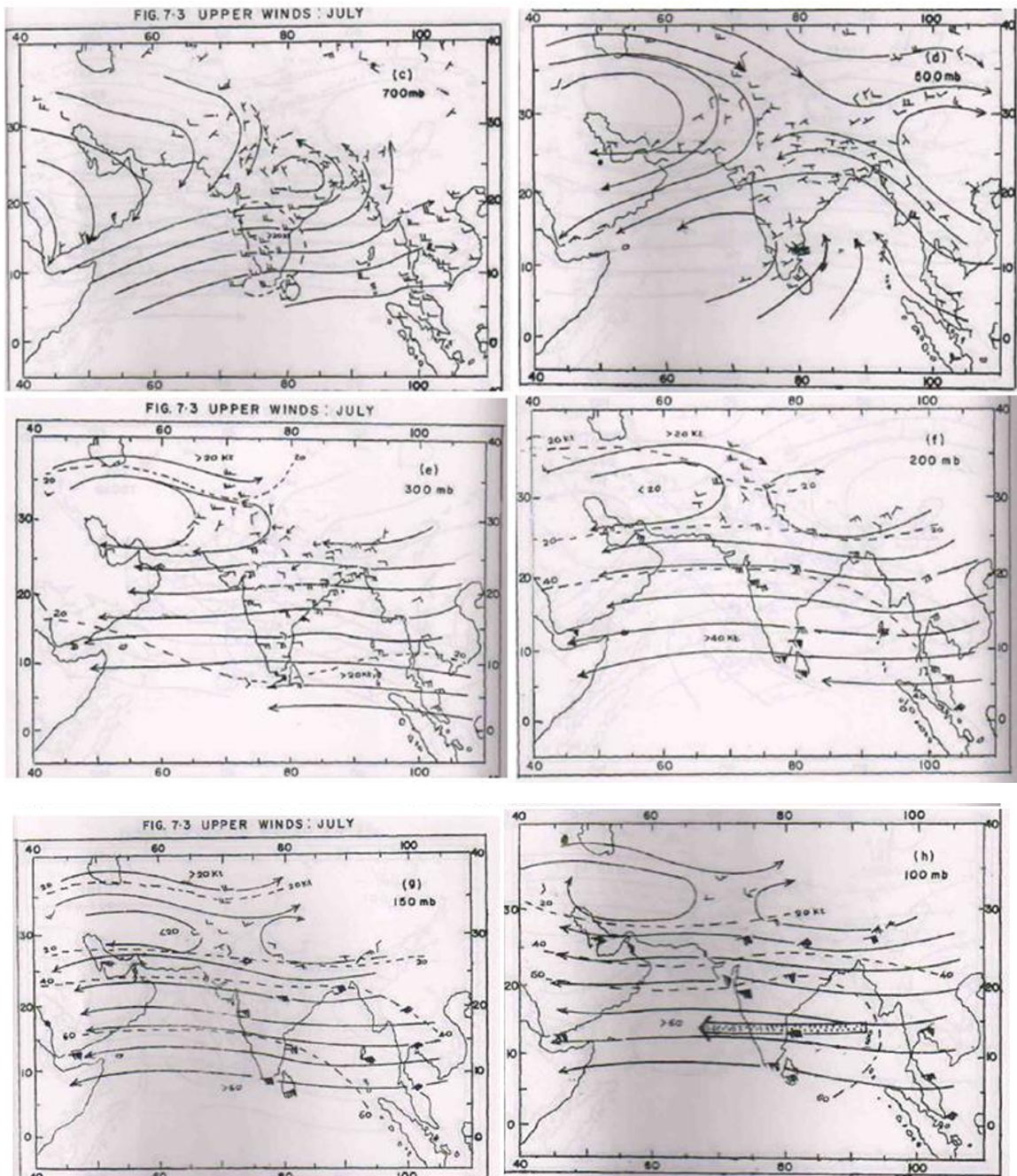
The rainfall gradually decreases from the east to the west It is less than 20 cm over western parts of West Rajasthan.

- The distribution of rainfall in India also depends on the position of the monsoon trough, when the monsoon trough gets stuck to the north of its original position i.e. the foothills of the Himalayas, resulting in a drastic reduction in rainfall over the mainland part of the country except for the foothills. this condition is called **Break Monsoon**.
- During Break, monsoon pressure rises over most parts of the country. During this period heavy rainfall occurs in the foothills of the Himalayas, mainly in West Bengal,

Nepal, and Bhutan resulting in floods over NE and Bihar. Surface wind becomes strong in this region.

— Flying condition: Relatively smooth in the stratiform cloud but turbulent in the CU/CB cloud. Visibility is good except during precipitation.

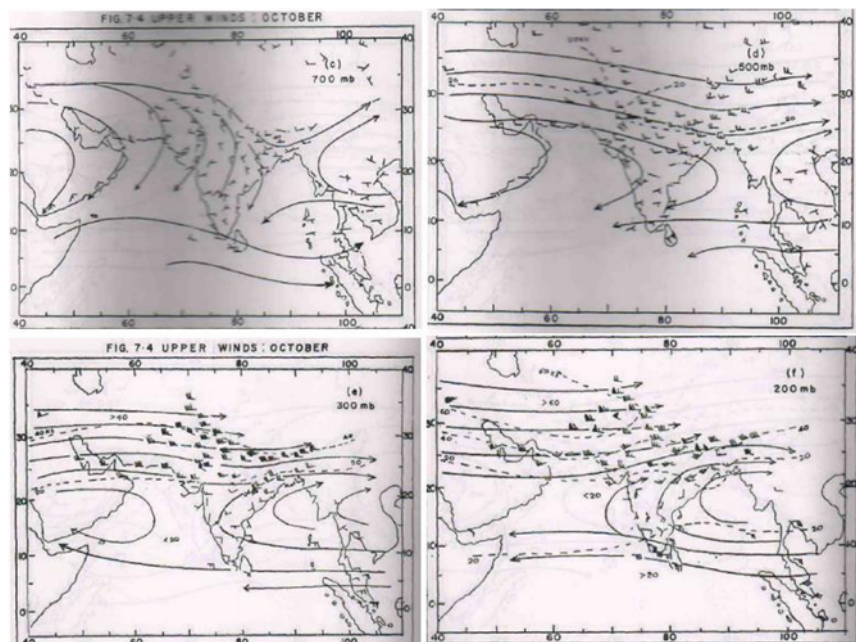
— Tropical Easterly Jet stream lies between 10° to 15° N i.e., close to Bangalore & Hyderabad, at a height of 15-16km (100 hPa) with a wind speed of 70-80 kts.

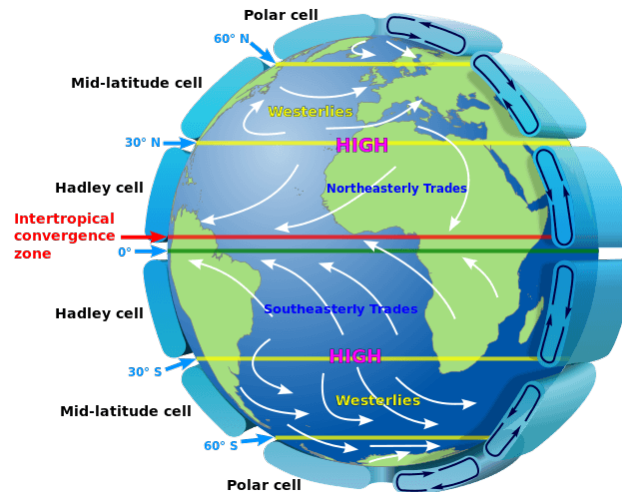


POST MONSOON (Transition) – October to November

Post monsoon season is the transition period from summer monsoon circulation to winter circulation. The temperature starts decreasing & Pressure starts Increasing, and the Pressure gradient starts decreasing.

- Post monsoon is also known as **Retreating Monsoon season**. The SW monsoon starts withdrawing from N.W India by September, withdraws south of peninsula till 15 Oct. **Southern part of India and Kerala experience rainfall during post monsoon season.**
- The pressure pattern is defused and ill-defined. Only 1 to 2 isobars can be drawn over entire India. The monsoon trough shifts to the Bay of Bengal along 13N. A weak low form over the Bay of Bengal adjoining the Chennai coast and another low is over Baluchistan.
- Rainfall activity over Andhra Pradesh coast and Tamil Nadu due to Thunderstorm activity over the Bay of Bengal.
- In the post monsoon season precipitation is highest (30 – 100 cm) in the south peninsula. East coast, Assam and adjacent states and Kashmir also receive considerable precipitation (15-20 cm) during the season. There is hardly any rainfall over Rajasthan, Haryana and Saurashtra and Kutch during the season.
- **The month of October has more number of storms and depressions and the December has the least.** Maximum number of storms develops during the month of November. The number of cyclonic storm and severe cyclonic storms is more in the first half of November than in the later half. Thus, **earlier half of the post monsoon season constitutes major storm season for India.**
- **October is the month of more frequent cyclonic storms in the Bay of Bengal & mainly hit Tamil Nadu, Andhra Pradesh, and Orissa coast on a westerly track.** Some of the storms move in a north or north-easterly direction.
- **Maximum thunderstorm activity occurs in India during Post monsoon.**
- The sub-tropical jet stream starts to appear over J&K, CAT may be experienced in north India.



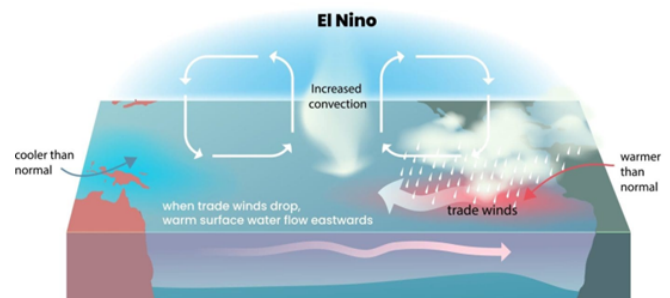


El Niño and La Niña

Pacific Ocean is this large pool of water that exists on the western side of the American continent and eastern side of the Asian and Australian continent this large pool of water gets intensely warm due to the fact that it exists right at the equator and we all know that the equatorial region received great amount of sun rays all throughout the year.

- ENSO stands for El Niño-Southern Oscillation, primarily occurring in the southern hemisphere, particularly in the Pacific Ocean.
- It involves the oscillation of ocean temperatures, shifting between warm and cool temperatures, affecting weather patterns globally.
- The process involves two main phases: El Niño, and La Niña, each characterized by specific oceanic and atmospheric conditions.

- During El Niño, weakened trade winds allow warm ocean currents to move towards the central and eastern Pacific, disrupting weather patterns and causing phenomena like droughts and floods.



- La Niña, on the other hand, is marked by strong trade winds, pushing warm ocean currents towards the western Pacific, leading to cooler ocean temperatures and distinct weather patterns, particularly affecting countries like Australia, Indonesia, and New Zealand.

